

The Data End of Crypto Quant for Public LLMs: An Observation Layer that Compiles World Bundles

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Abstract

Purpose. End-to-end crypto quant for public LLMs needs a *data end* before decision or execution. We build that layer: compile asynchronous multi-band evidence into a point-in-time (PIT) *world bundle*, and hard-refuse when support is thin so half-baked context cannot drive downstream decisions. **Evidence.** Primary—seeing: grounding recovers ready/missing/tilt at ≈ 0.97 under Compiled vs. $\lesssim 0.23$ under Raw. Secondary—valve: Compiled thin-abstain is 1.0 across four vendors while Ungated soft judgment on the same content is only ≈ 0.68 ; Raw over-trades. Secondary—content: vintaged tilts beat momentum under a no-LLM rule ($\Delta CE = 0.334$, $p = 0.034$). The panel is scarce (max WMI $\approx 0.093 < 0.2$), so production Compiled never opens—a valve stress test. **Claim.** Build the quant data end first: compile so models can see; refuse when they must not act.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

data end, observation layer, world bundle, grounding, public LLMs, cryptocurrency, point-in-time, quality-gated refusal, systems

ACM Reference Format:

Anonymous Author(s). 2026. The Data End of Crypto Quant for Public LLMs: An Observation Layer that Compiles World Bundles. In *Proceedings of the 7th ACM International Conference on AI in Finance (ICAIF '26)*. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/XXXXXXX.XXXXXXX>

1 Introduction

Purpose (read this first). Build the *data end* of end-to-end crypto quant for public LLMs. The stack is **data/observation** → **decision** → **execution**; agents and generative world models [6, 7, 9, 19–21] typically assume the first stage. Crypto violates that assumption: exchange, macro, and alternative evidence arrive on incompatible clocks [11, 13]. We own only the data end—compile a PIT world bundle so models can *see*, and hard-refuse when the view is too thin. Refusal is a safety valve, not the product and not a trading claim.

Failure modes. No feed → refuse (safe, useless). Thin price fragment → trade and lose. Disclosed but ungated world → read fields, disagree on action. So the data end must (i) make the market legible and (ii) enforce a quality valve before decision/execution.

What we build. An anonymized runtime (Section 4): multi-band evidence → BandPIT/WMI → JSON world bundle for GPT/DeepSeek/GLM/decision-class models. “World model” means state compiler + quality gate

(Section 3), not dynamics and not a decision engine. The bundle is complete, honest, and auditable; `should_ai_abstain` is the machine-readable valve.

How we evaluate. **RQ1 (primary—seeing):** Are ready /missing /tilt answers verifiable against PIT ground truth under Compiled vs. Raw? **RQ2 (valve):** Does a hard flag enforce thin abstention where soft disclosure does not? Four arms: Compiled, Ungated (same content, no flag), Raw (mom5), Blind. **RQ3 (secondary):** Are vintaged tilts nonempty under a transparent no-LLM rule?

Contributions. (1) Data-end semantics: epistemic observations, Π_t , and seeing vs. gating. (2) Anonymized PIT compile path: BandPIT previous-close clock, quality-tagged bundles, frozen four-arm harness. (3) Live + rule validation on a scarce three-band archive: grounding ≈ 0.97 ; thin-abstain 1.0 vs. Ungated ≈ 0.68 ; $\Delta CE = 0.334$. Estimand: legibility and valve reliability—not decision SOTA, execution, or Compiled-open alpha.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

LLM agents in finance. Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how agents fetch, debate, and are scored. They still assume a usable observation stream. We study that prior stage: compile a world bundle, measure whether models can *see* it, then gate action when they must not act.

Selective prediction and PIT. Abstention can be optimal under noise [4, 5]; we attach refusal to world quality via a runtime field, not classifier confidence alone. Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation. Bootstrap discipline for the content probe follows [8, 14, 17].

3 Observation Layer

Definition 3.1 (Epistemic observation). $O_{j,t} = (x, \tau, q, g, r)$: value, latest-available time, quality, main-view gate, semantic role. Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (WMI/ACWMI); $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$. Π_t aligns clocks, gates honesty/missingness, and emits consumer JSON—the handoff to decision.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $WMI_t = B_t U_t H_t$ aggregates breadth, stability, and honesty; ACWMI adds regime weight. Compiled exposes `should_ai_abstain` and `thin_world`.

Table 1: Compile-path modules (not a product inventory).

Module	Role
Collectors / stores	Timestamped / vintaged band history
BandPIT compiler	Align clocks; ready/stale/missing; Π_t
Quality indices	$B, U, H \rightarrow$ WMI; abstain flag
Bundle builder	Complete / honest / auditable JSON
Consumer harness	Four-arm protocol; temp. 0

Table 2: Input evidence vs. fields emitted in the world bundle.

Band	Fetches (raw)	Emitted (bundle)
exchange	OHLCV, funding, OI	mom5; status/age
macro	equity-vol / dollar indices	macro_tilt; status/age
alternative	stablecoin circulation	alt_tilt; status/age
derived	—	WMI, abstain flag, evidence IDs

Seeing vs. gating. The bundle is what the model *sees*; the hard flag *blocks* action when $q(W_t)$ is too low. Tilts without a gate are a dump; a gate without a readable bundle is only a switch.

PROPOSITION 3.3 (COMPILATION $\neq$ DUMP). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{AI}$: ungated stale evidence can raise volume while lowering honesty.*

PROPOSITION 3.4 (WORLD-CONDITIONAL ABSTENTION). *If non-abstain actions exceed a world-dependent cost, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

REMARK 3.5 (NOT A GENERATIVE WM). *We do not learn $p(s_{t+1} | s_t, a_t)$. We deliver a state compiler with a quality-gated refuse interface.*

4 Runtime: Compile Path

Figure 1 and Table 1 summarize the anonymized data-end prototype: collectors write multi-band history; BandPIT compiles a PIT state; a service emits a JSON world bundle; frozen-prompt consumers read it at temperature 0. Decision and execution sit downstream; this paper does not claim them.

What enters the view. This paper evaluates three bands that populate the consumer object (Table 2). *Exchange*: OHLCV klines, funding, open interest. *Macro*: equity-volatility and dollar indices with available_at vintages. *Alternative*: aggregate stablecoin circulation. Every row carries an observation timestamp so compilation can enforce a previous-close information set. The schema is band-extensible; the experiments below score the view built from these three feeds.

How compilation builds the view. For calendar day t , BandPIT reconstructs the decision information set at $(t-1)$ 23:59 (payoff is same-day close-to-close return). Each band is graded ready / limited / missing against age thresholds; missingness is disclosed rather than imputed. Breadth, stability, and honesty aggregate into $WMI_t = B_t U_t H_t$ (ACWMI adds a regime weight). PIT-safe tilts use only in-clock observations: five-day macro changes, a seven-day

Table 3: World-bundle field groups (the market view).

Group	Fields
Timing	decision_asof, previous-close protocol
Completeness	ready/limited/missing counts, band status
Quality	WMI, ACWMI, abstain flag, thin_world
Content	macro_tilt, alt_tilt, mom5
Audit	evidence_ids

stablecoin-flow slope, and mom5. The valve sets `should_ai_abstain` when $WMI < 0.2$ and exposes `thin_world`—a switch on the view, not a substitute for seeing. Figure 2: macro and alternative stay ready; exchange readiness varies and defines thick/thin days.

What the LLM sees: the world bundle. The consumer object is a JSON world bundle (Table 3, Listing 1): timing, completeness, quality (including the abstain flag), content tilts, and evidence IDs that bind rationales to disclosed bands. That package is the market view we ask public LLMs to read. Outputs used in experiments are a $\sim 3,990$ -row PIT panel (10 assets $\times$ 399 days; 2025-07 to 2026-08) and the per-day bundles fed to the four arms.

Listing 1: World bundle example (what Compiled sees).

```
{
  "date": "2026-05-11",
  "asset": "ADA",
  "decision_asof": "2026-05-10T23:59:00",
  "completeness": {
    "n_ready": 2,
    "n_missing": 1
  },
  "world_model_index": {
    "wmi": 0.015,
    "should_ai_abstain": true,
    "thin_world": true
  },
  "macro_tilt": 1.0,
  "alt_tilt": -1.0,
  "audit": {
    "evidence_ids": [
      "band:exchange:missing",
      "band:macro:ready"
    ]
  }
}
```

How consumers use the view. **Compiled**: bundle + hard abstain when the quality flag is true. **Ungated**: same content and numeric WMI; soft “you may abstain.” **Raw**: mom5 only. **Blind**: no feed. max WMI ≈ 0.093 , so production Compiled abstains on every OOS day—a valve stress test; RQ1 still scores legibility of the disclosed view. Compiled vs. Ungated differ only in enforcement (Listings 2–3); prompts, IS/OOS cut, and seeds are frozen.

Listing 2: Compiled (excerpt): hard valve.

```
1. If should_ai_abstain is true OR the world is thin,
   you MUST "abstain".
2. Prefer band content over raw momentum.
3. Rationale must match evidence_ids.
```

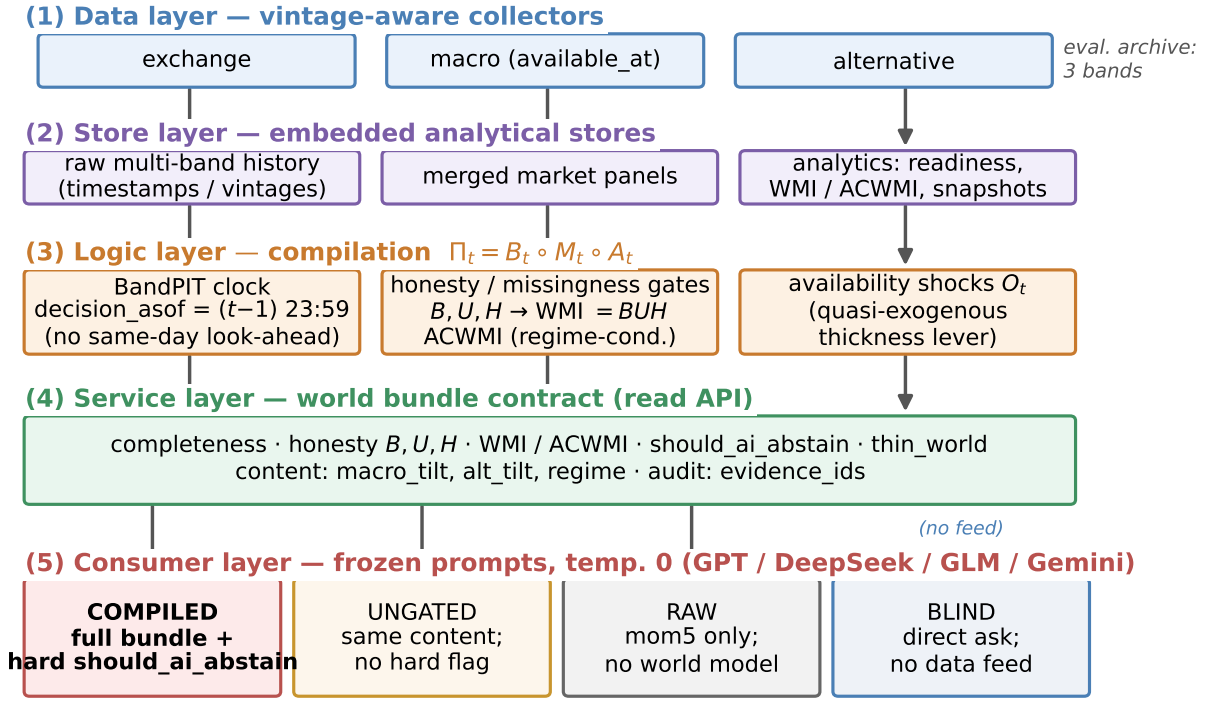
Listing 3: Ungated (excerpt): soft judgment.

```
1. Judge whether low WMI / missingness makes action
   unsafe; you MAY "abstain".
2. Prefer band content when you act.
   NO hard abstain boolean; NO forced policy.
```

5 Experiments

5.1 Setup

PIT panel ~ 399 days, 10 liquid names, chronological IS/OOS 200/200 days. Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors. The no-LLM probe uses 199 OOS days after return alignment. Primary metrics for RQ1 are grounding F1/accuracy; for RQ2, thin-world abstain rate (CE as avoided loss, not alpha).



Live abstention outcomes: Sec.~Experiments (Table / four-arm figure).

Figure 1: Data-end pipeline: evidence → BandPIT → world bundle → Compiled / Ungated / Raw / Blind.

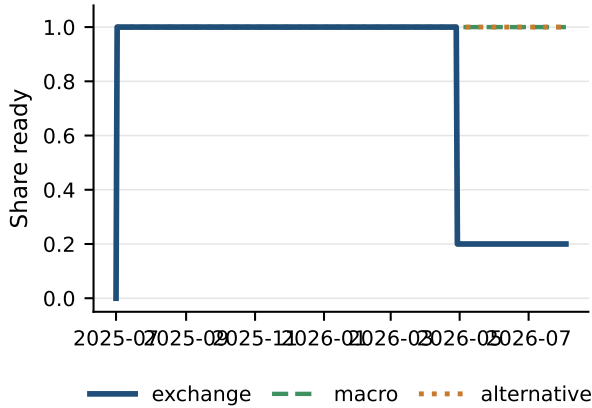


Figure 2: PIT readiness: what is visible in the archive over time.

5.2 RQ1 (primary): Seeing—grounding workflow

If the data end works, models must report a *checkable* market view—not invent coverage from a price fragment.

On 50 OOS asset-days, each model lists ready/missing bands among exchange/macro/alternative, reports macro_tilt / alt_tilt

Table 4: Grounding (50 days): primary seeing result. C = Compiled bundle; R = Raw.

Model	Ready F1		Missing F1		Tilt-sign	
	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

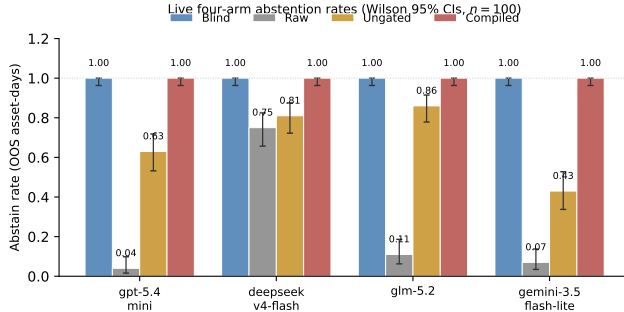
signs, and judges sufficiency; (i)–(iii) score against PIT ground truth.

Table 4: Compiled mean ready/missing F1 and tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. Raw has no readiness or tilt fields, so it cannot recover archive state; Compiled forces use of the disclosed world. Sufficiency chat alone is weak (models may call 3/3-ready “enough” while WMI marks thin), so we separate legibility (RQ1) from the valve (RQ2).

Case study (same asset-day). Listings 4–5 excerpt live rationales on ADA (2026-05-11): exchange missing; macro/alternative ready; thin world. Compiled recovers the checkable view and judges

Table 5: Thin-world abstention on identical 100 OOS days (Wilson 95% CIs). Blind = 1.0.

Model	Thin C	Thin U [CI]	Abs. R [CI]	Δ CE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

**Figure 3: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled enforces the hard gate.**

the world insufficient. Raw cannot list readiness or tilt signs because the fragment has no world fields. A second pair on ADA (2026-01-17) shows the valve contrast: Compiled abstains because `should_ai_abstain` is true; Raw goes bullish from `mom5` = +1 alone—half-baked context driving action.

Listing 4: Compiled grounding (gpt-5.4-mini, ADA 2026-05-11).

```
ready: alternative, macro
missing: exchange
macro_tilt +; alt_tilt -; sufficiency: insufficient
"thin world... should_ai_abstain=true"
```

Listing 5: Raw grounding / action (same model family).

```
grounding: ready=[]; cannot confirm tilts
action (2026-01-17): bullish
"Positive 5-period momentum in raw feed."
```

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. Table 5 and Figure 3: on the same 100 days, Compiled abstains on every thin day (1.0); Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 6); Blind abstains 1.0 with no feed. The costly failure mode is the middle case: half-baked Raw context makes models act and lose. Mean Δ CE (Compiled–Raw) +1.07 is avoided loss under the valve, not an alpha claim. Full-OOS ($n=2000$, all thin) reproduces Compiled 1.0 and Ungated mean ≈ 0.56 .

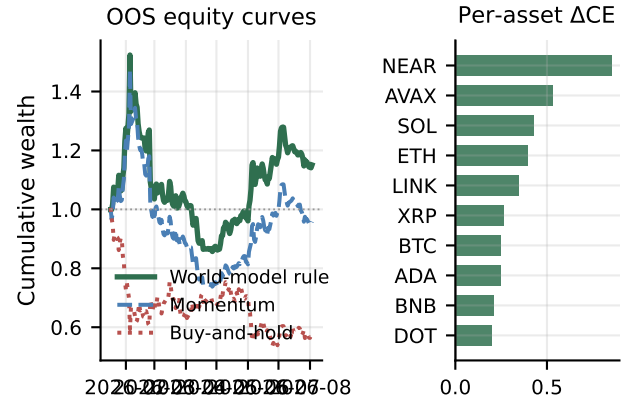
Open slice (content lower bound). Production gate 0.2 never opens on this panel (max WMI ≈ 0.093); $WMI \geq 0.05$ selects the band-thick subset ($\approx 61\%$ OOS; $n \approx 1224$). On that slice Ungated (same

Table 6: Raw action mix (100 days): half-baked context drives directional mass.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Table 7: No-LLM content probe (secondary). Bootstrap Δ CE 0.334; raw gap 0.336.

Policy	Sharpe	CE
Buy-and-hold	−1.404	−1.163
Momentum	0.096	−0.206
World-bundle tilts	0.764	+0.130
Tilts – Momentum	—	0.334 ($p=0.034$)

**Figure 4: Secondary content probe: compiled tilts beat momentum; Δ CE positive for every asset.**

content, no hard boolean) has mean abs. ≈ 0.48 and CE positive for every vendor (mean $\approx +0.29$; bootstrap $\approx +0.31$, CIs include 0)—a field lower bound, not Compiled-open. Production Compiled still abstains at 1.0; Raw stays risky (GPT/Gemini abs. 0.0, CE negative).

5.4 RQ3 (secondary): Nonempty bundle fields

A transparent no-LLM rule on vintaged `macro_tilt` / `alt_tilt` beats momentum on shared returns (Table 7, Figure 4): OOS Sharpe 0.764 / CE +0.130 vs. momentum (0.096, −0.206); block-bootstrap Δ CE=0.334 ($p=0.034$). Leave-one-band-out attributes the gap to content, not the refuse flag; a non-vintaged audit shows no edge. On the open/thick slice Δ CE=0.315; on exchange-gap days the rule abstains more (Δ CE=0.129)—complementary to the valve test, not a thin-day profit claim.

5.5 Discussion

What carries the paper. (i) Legible data end (RQ1 grounding). (ii) Hard valve where soft text fails (RQ2). (iii) Nonempty fields

under a no-LLM probe and open-slice lower bound (RQ3). Relative to FinMem /FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we supply the observation stage those systems assume—not a competing decision or execution layer.

Why grounding first. Trading PnL confounds observation quality with policy skill. Grounding asks whether the consumer can audit ready/missing/signed state; Raw cannot, Compiled can. The valve then asks whether soft judgment stops action on a thin disclosed world; empirically, no.

Design. World bundle not feature dump; previous-close PIT; four arms separate seeing+gating, seeing without gating, thin wiring, and no feed. Frozen prompts, temperature 0, IS/OOS cut, and seeds; anonymized harness on acceptance.

6 Conclusion

Public LLMs need a data end before they decide. We compile asynchronous crypto evidence into a PIT world bundle and hard-refuse when the view is too thin. Grounding ≈ 0.97 vs. Raw collapse; thin-abstain 1.0 where soft judgment fails; vintaged tilts nonempty ($\Delta CE=0.334$). Build the quant data end first: compile so models can see; refuse when they must not act.

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